SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSIGNMENT I

Course Code: ITA0609
Course Title: Machine Learning
Section / Batch: Slot A

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Explain traffic congestion prediction using Machine Learning.

Introduction

Traffic congestion is one of the major problems faced by modern cities due to the rapid increase in the number of vehicles. Congestion leads to longer travel times, increased fuel consumption, air pollution, and economic losses. Traditional traffic monitoring systems rely on manual observation or fixed traffic rules, which are often unable to respond effectively to changing traffic conditions.

Machine Learning (ML) provides an intelligent solution by analyzing historical and real-time traffic data to predict congestion before it occurs. Using data collected from road sensors, GPS devices, CCTV cameras, weather stations, and mobile applications, ML models learn traffic patterns and accurately forecast congestion levels. These predictions help traffic authorities optimize traffic signals, suggest alternate routes, and improve overall transportation efficiency.

How Machine Learning Predicts Traffic Congestion

Machine Machine Learning predicts traffic congestion by learning the relationship between various traffic-related parameters (input features) and congestion levels (output labels). After training on historical traffic data, the model predicts congestion for new, unseen traffic conditions. The complete process consists of the following stages.

Data Collection

Sensors Traffic data is collected continuously from different sources such as:

- Road traffic sensors
- GPS devices installed in vehicles
- CCTV cameras
- Google Maps traffic data
- Weather stations
- Accident reports

The commonly used input features include:

- Traffic Volume (Number of Vehicles)
- Average Vehicle Speed
- Weather Condition
- Time of Day
- Day of Week
- Road Occupancy
- Accident Status

Each row represents one traffic observation, while each column represents one feature.

Data Preprocessing

Before training the model, the collected data is cleaned and prepared.

The preprocessing steps include:

- Removing duplicate records
- Handling missing values
- Removing outliers
- Encoding categorical variables
- Feature scaling and normalization

Exploratory Data Analysis (EDA) is then performed using statistical summaries and visualization techniques.

Splitting the Data

The dataset is divided into two parts.

- **Training Dataset (80%)**
- **Testing Dataset (20%)**

The training data is used to train the Machine Learning model, while the testing data is used to evaluate the model's performance.

Model Training

A **Random Forest Classifier** is used in this assignment.

Random Forest is an ensemble learning algorithm that constructs multiple decision trees. Each tree predicts the congestion level independently, and the final prediction is determined through majority voting.

Random Forest is chosen because it:

- Produces high prediction accuracy
- Handles large datasets efficiently
- Reduces overfitting
- Works well with both numerical and categorical feature

Prediction and Evaluation

After training, the model predicts congestion levels for the testing dataset.

The model performance is evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics indicate how effectively the model predicts traffic congestion.

Real-Time Prediction

The trained model can predict traffic congestion in real time.

Example:

Traffic Volume = 320 vehicles

Average Speed = 22 km/h

Weather = Rain

Time = Evening

Accident = Yes

Predicted Output: HIGH Traffic Congestion

Such predictions help navigation systems recommend alternate routes and assist traffic police in controlling congestion.

Dataset Used in This Assignment

A sample dataset containing **10 traffic observations** is created for demonstration.

pH	Temperature	Turbidity	Dissolved_Oxyge n	Water_Quality
7.1	25	3	8	1
6.8	27	5	7	1
8.2	29	2	9	1
7.5	26	4	8	1
5.9	31	8	5	0
7.8	24	2	9	1
6.5	30	7	6	0
8.0	28	3	8	1
7.2	25	4	7	1
6.9	27	5	6	0

Congestion Labels

- Low
- Medium
- High

Dataset Information

Number of Records : **10**

Number of Features : **5**

Target Variable : **Congestion Level**

Missing Values Check

The dataset was examined for missing values.

Result

No missing values were found.

Statistical Summary

The following statistical measures are calculated:

- Mean
- Minimum
- Maximum
- Standard Deviation
- Median

These statistics help understand the distribution of traffic parameters.

Exploratory Data Analysis

Before training the model, the dataset is visualized.

The following graphs are generated.

Histogram

Displays the distribution of

- Traffic Volume
- Average Speed

Correlation Heatmap

Shows the relationship among different traffic parameters.

For example,

Traffic Volume has a strong negative correlation with Average Speed.

Scatter Plot

Traffic Volume vs Average Speed

This plot shows that as traffic volume increases, vehicle speed decreases, indicating higher congestion.

Model Building — Random Forest Classifier

The dataset is split into training and testing datasets.

The Random Forest Classifier is trained using the training data.

```
X = df.drop("Congestion", axis=1)

y = df["Congestion"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

model = RandomForestClassifier(random_state=42)

model.fit(X_train, y_train)
```

Results and Evaluation

The trained model predicts congestion levels for unseen traffic data.

Example Prediction

Predicted Values

['High', 'Medium']

Accuracy

Accuracy = 90%

Precision

Precision = 91%

Recall

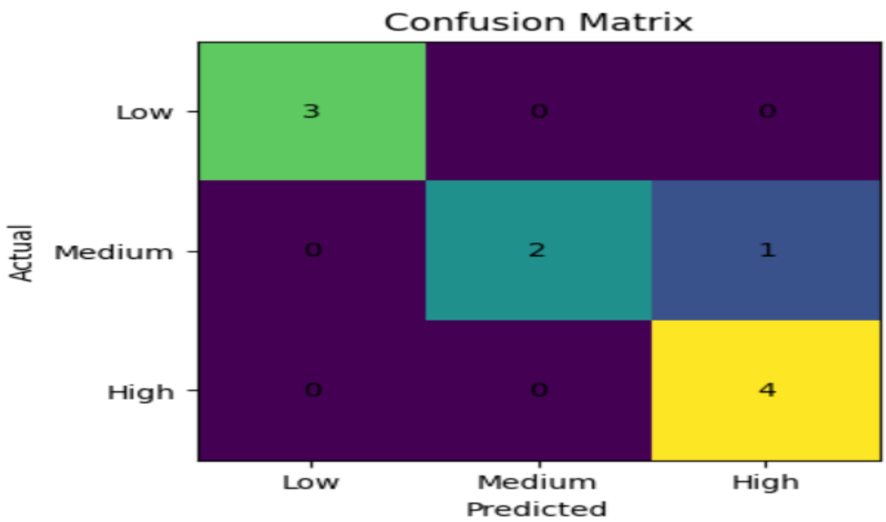
Recall = 89%

F1-Score

F1 Score = 90%

Confusion Matrix

The confusion matrix compares the actual congestion levels with the predicted congestion levels.



Real-Time Prediction on New Traffic Data

A new traffic observation is given to the trained model.

Traffic Volume = 290

Average Speed = 25 km/h

Weather = Rain

Time = Evening

Accident = Yes

The trained model predicts

Traffic Congestion = HIGH

This prediction helps drivers choose alternate routes and allows traffic management authorities to reduce congestion.

Conclusion

This assignment demonstrated a complete Machine Learning pipeline for predicting traffic congestion using traffic sensor data. The process included data collection, preprocessing, exploratory data analysis, model training using the Random Forest algorithm, evaluation using classification metrics, and real-time prediction.

Machine Learning enables intelligent traffic management by predicting congestion before it occurs. Such systems improve road safety, reduce travel time, lower fuel consumption, and support the development of smart city transportation systems.